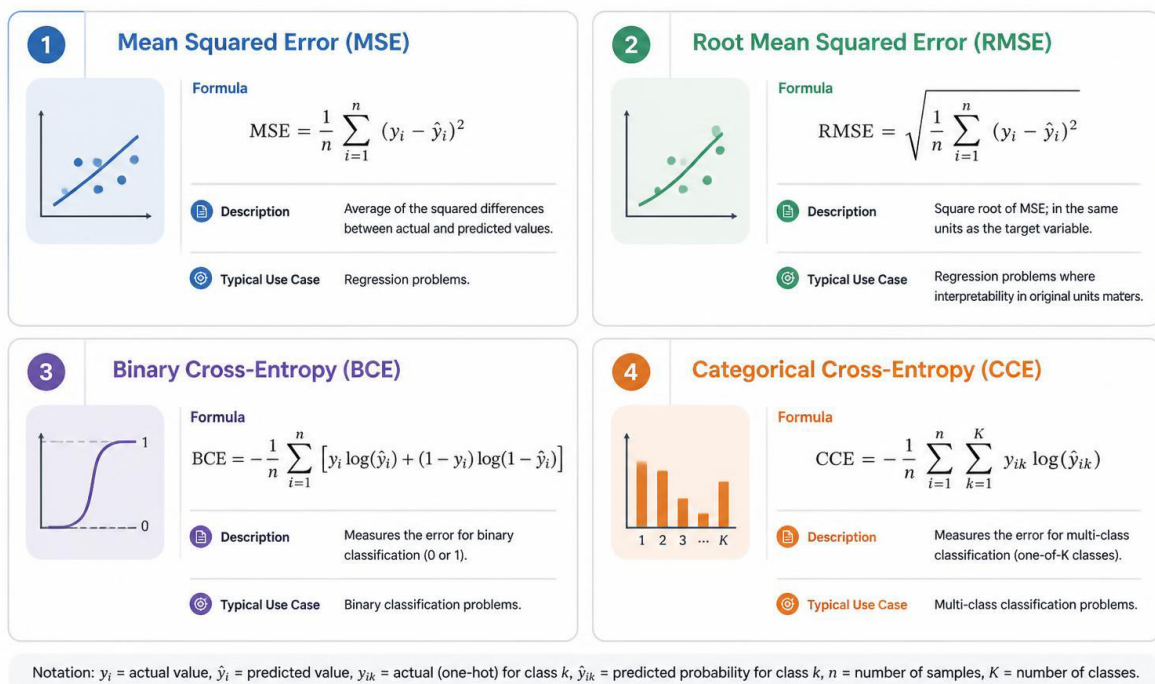


Some of the frequently used cost functions for Machine Learning:

- MSE (Mean Squared Error)
- RMSE (Root Mean Squared Error)
- binary cross-entropy
- categorical cross-entropy

Two of them are explained in comprehensive format below for quick revision. Enjoy “the echo” effect!

4 Common Machine Learning Loss Functions



MSE = Mean Squared Error

the average of the squared differences between actual and predicted values.

- Use: Regression loss/metric; penalizes larger errors more than MAE.
- Interpretation: Units are squared units of the target (not directly comparable to target scale).
- Formula:

$$\text{MSE} = (1/n) * \sum (y_i - \hat{y}_i)^2$$

where y_i = actual, $\hat{y}_i$ = predicted, n = number of observations.

- Relationship to RMSE: $\text{RMSE} = \sqrt{\text{MSE}}$. Code fragment:

```
import numpy as np  
  
mse = np.mean((y_true - y_pred)**2)
```

RMSE = Root Mean Squared Error

a common metric for measuring the average magnitude of prediction errors. Brief summary and formula:

- Use: Evaluate regression models; lower is better.
- Interpretation: Same units as target variable; sensitive to large errors (squares penalize outliers).
- Formula:

$$\text{RMSE} = \sqrt{\frac{1}{n} * \sum (y_i - \hat{y}_i)^2}$$

where y_i = actual, $\hat{y}_i$ = predicted, n = number of observations.

Quick notes:

- Compare RMSE to the data's standard deviation or range to judge usefulness.
- For model selection, pair with other metrics (MAE, R^2). Code fragment:

```
import numpy as np

rmse = np.sqrt(np.mean((y_true - y_pred)**2))
```

Mark your feedback at the Following Contact Points

- **LinkedIn:** <<https://www.linkedin.com/in/rishirajopenminds>>
- **GitHub:** <<https://github.com/rishiraj88>>

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